

User Microservice Specification - MiniEdu

Tech Stack

- Java 17+
- Spring Boot 3+
- Spring Security + JWT
- Spring Data JPA
- Hibernate Validator (Bean Validation)
- PostgreSQL (or H2 for dev)
- Lombok (optional)
- Java Records for DTOs
- Docker (for containerizing)

Project Structure

com.miniedu.userservice

controller

dto

entity

repository

security

service

impl

exception

config

User Entity

Fields:

- UUID id
- String username
- String email

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- String password
- UserRole role (enum: USER, ADMIN)
- LocalDateTime createdAt

DTOs (record types)

- RegisterRequest(String username, String email, String password)
- LoginRequest(String email, String password)
- UserResponse(UUID id, String username, String email, String role)
- AuthResponse(String token, UserResponse user)

Service Methods

- UserResponse register(RegisterRequest request)
- AuthResponse login(LoginRequest request)
- UserResponse getCurrentUser(String token)
- boolean existsByEmail(String email)
- Optional<User> findByEmail(String email)

Security Layer Components

- JwtTokenProvider: generate/validate JWT
- JwtAuthenticationFilter: intercept requests and check token
- CustomUserDetailsService: load user from DB for Spring Security
- SecurityConfig: configure auth rules, password encoder, etc.

AuthController Endpoints

Endpoint	Method	Description
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/api/auth/register	POST	Register new user

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/api/auth/login	POST	Login, return JWT	
/api/auth/me	GET	Get authenticated user	

UserRepository Methods

- Optional<User> findByEmail(String email)
- boolean existsByEmail(String email)

Exception Handling

- GlobalExceptionHandler with @ControllerAdvice
- Custom exceptions:
 - EmailAlreadyExistsException
 - InvalidCredentialsException
 - UserNotFoundException

Security Details

- Use BCryptPasswordEncoder for password hashing
- JWT is stateless, no need for session
- Secure routes with @PreAuthorize if needed
- Add roles as GrantedAuthority during authentication

Validation

- Use @Valid in controller methods
- Annotate record fields with:
 - @NotBlank
 - @Email
 - @Size

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Optional Features

- Swagger/OpenAPI (e.g., springdoc-openapi)
- Dockerfile
- Integration tests with Testcontainers